

*We evaluate the out-of-sample performance of the sample-based mean-variance model, and its extensions designed to reduce estimation error, relative to the naive 1/N portfolio. Of the 14 models we evaluate across seven empirical datasets, none is consistently better than the 1/N rule in terms of Sharpe ratio, certainty-equivalent return, or turnover, which indicates that, out of sample, the gain from optimal diversification is more than offset by estimation error.*⁹

The DeMiguel, Garlappi, and Uppal paper perplexed the academic establishment and drove home one of the key takeaways of successful asset allocation: *Complexity does not necessarily add value.*¹⁰ Simply stated, it is hard, and perhaps impossible, to do much better than 1/N in an hypercompetitive world with so much volatility and uncertainty.

But Which Assets Do We Choose?

Research suggests that an equal-weight approach works. It also seems reasonable that we approach the world as the endowments do and envision three core buckets: equity, real assets, and bonds. We could include alternatives in our setup, but this asset class can get messy and is generally terrible when we assess the FACTS outlined in the previous chapter. Let's keep the conversation practical for the do-it-yourself financial advisor.

How should we think about equity, real assets, and bonds? We believe that Mebane Faber and Eric Richardson's Ivy 5 portfolio achieves our goal with minimal brain damage ([Figure 6.2](#)).¹¹ They envision the world with five core asset classes, all equal-weighted at 20 percent:

- *Equity (40%)*

SP500 = SP500 Total Return Index (US stocks)

EAFE = MSCI EAFE Total Return Index (international stocks)

- *Real assets (40%)*

REIT = FTSE NAREIT All Equity REITS Total Return Index (real estate)

GSCI = Goldman Sachs Commodity Index (commodities)